

THE SHADOW OBSTRUCTION TOWER OF A CHIRAL ALGEBRA IS THE L_∞ -FORMALITY OBSTRUCTION TOWER

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ABSTRACT. Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve and let $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ be its modular convolution dg Lie algebra, with bar-intrinsic Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$. We prove that the shadow obstruction tower $(\text{Sh}_r(\mathcal{A}))_{r \geq 2}$, defined as the finite-order projections of $\Theta_{\mathcal{A}}$ under the bar filtration, coincides with the L_∞ -formality obstruction tower of the genus-zero part $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(0)}$. The identification is established at all degrees by induction from explicit base cases at $r = 2, 3, 4$. As a corollary the operadic complexity conjecture is proved: the shadow termination degree, the transferred A_∞ -depth, and the L_∞ -formality level of $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(0)}$ are a single invariant of the chiral algebra.

1. INTRODUCTION

1.1. Formality and its obstructions. A dg Lie algebra $\mathfrak{g}$ is L_∞ -formal when its minimal model, obtained by homotopy transfer onto cohomology, carries trivial higher brackets $\ell_n = 0$ for $n \geq 3$; equivalently, $\mathfrak{g}$ is L_∞ quasi-isomorphic to $(H^*(\mathfrak{g}), 0, [-, -])$. Kontsevich’s formality theorem [3] proved that polyvector fields on a smooth manifold form an L_∞ -formal dg Lie algebra, giving deformation quantization of Poisson structures; Tamarkin [8] recovered the result operadically from formality of the little disks. When formality fails, obstructions are recorded in cohomology classes $o_3, o_4, \dots$: at degree r the class o_r measures whether ℓ_r^{tr} vanishes on cohomology; if $o_r = 0$, a homotopy kills ℓ_r^{tr} and the next obstruction lives one degree higher. The sequence $(o_r)_{r \geq 3}$ is the *formality obstruction tower*; see Loday–Vallette [5, Ch. 10].

1.2. The shadow tower of a chiral algebra. On a smooth projective curve X , a chiral algebra $\mathcal{A}$ [1] has a bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, where $\bar{\mathcal{A}} = \ker(\epsilon)$ is the augmentation ideal, with deconcatenation coproduct and differential built from the Arnold form $\eta_{ij} = d \log(z_i - z_j)$ on the Fulton–MacPherson compactification of the configuration space. The modular convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ is the pronilpotent dg Lie algebra whose differential encodes graph sums over stable curves in $\bar{\mathcal{M}}_{g,n}$. The bar-intrinsic element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$ satisfies $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ and packages the full genus expansion of $\mathcal{A}$ [6, Thm. H]. Its finite-degree truncations $\Theta_{\mathcal{A}}^{\leq r}$ define the *shadow obstruction tower*

$$\text{Sh}_r(\mathcal{A}) := -\nabla_H^{-1}([\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(0)}), \quad r \geq 2,$$

where the bracket is genus-zero graph composition and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\text{sh}}$. The first three shadows are $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ (the modular characteristic [6, Thm. D]), $\text{Sh}_3(\mathcal{A}) = \mathfrak{C}(\mathcal{A})$ (cubic shadow), and $\text{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A})$ (quartic shadow).

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1.3. The main theorem. The classical formality obstruction tower lives in Harrison cohomology of a dg Lie algebra; the shadow obstruction tower lives in the shadow algebra of a chiral algebra on a curve. Nothing in the definitions forces the two sequences to agree.

Theorem 1.1 (Shadow–formality identification; main theorem). *For every chiral algebra $\mathcal{A}$ and every degree $r \geq 2$,*

$$\mathrm{Sh}_r(\mathcal{A}) = \ell_r^{(\circ), \mathrm{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \dots, \Theta_{\mathcal{A}}^{\leq r-1}),$$

where $\ell_r^{(\circ), \mathrm{tr}}$ is the transferred genus-zero L_∞ bracket of degree r on the minimal model of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, (\circ)}$. Consequently the shadow obstruction tower of $\mathcal{A}$ is identical, as a sequence of cohomology classes, to the L_∞ -formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, (\circ)}$.

Theorem 1.1 is the self-contained extraction of the shadow–formality identification theorem in [6, Ch. on higher-genus modular Koszul]. The proof is by induction on r : the base cases $r = 2, 3, 4$ are verified explicitly (Proposition 4.1) and the inductive step uses the Kadeishvili–Merkulov tree formula [2, 7] on the strict genus-zero convolution algebra. The tree sum defining $\ell_r^{(\circ), \mathrm{tr}}$ and the graph sum defining Sh_r range over the same combinatorial data with matching propagators and coefficients.

1.4. Consequences. Theorem 1.1 resolves the operadic complexity conjecture: the shadow termination degree $r_{\max}(\mathcal{A})$, the transferred A_∞ -depth $d_\infty(\mathcal{A})$, and the L_∞ -formality level $f_\infty(\mathcal{A})$ of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, (\circ)}$ are a single invariant of $\mathcal{A}$ (Corollary ??). The four shadow-depth classes are realized by Heisenberg (class **G**, $r_{\max} = 2$, formal), affine Kac–Moody (class **L**, $r_{\max} = 3$), $\beta\gamma$ (class **C**, $r_{\max} = 4$), and Virasoro (class **M**, $r_{\max} = \infty$, intrinsically non-formal); see Section 7.

1.5. Organization. Section 2 defines the shadow obstruction tower. Section 3 recalls the transferred L_∞ structure and the formality obstruction tower. Section 4 proves the identification at degrees 2, 3, 4. Section 5 proves Theorem 1.1. Section 6 deduces the operadic complexity corollary. Section 7 verifies the four standard families.

Conventions. Grading is cohomological with $|d| = +1$; desuspension lowers degree, $|s^{-1}v| = |v| - 1$. Bar complexes use the augmentation ideal: $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The propagator $d \log E(z, w)$ on the bar complex has weight one. All statements are at the level of cohomology classes unless otherwise noted.

2. THE SHADOW OBSTRUCTION TOWER

2.1. The bar complex and its convolution algebra. Let X be a smooth projective curve over $\mathbb{C}$ and let $\mathcal{A}$ be a chiral algebra on X [1]. The ordered chiral bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}) = \bigoplus_{n \geq 0} (s^{-1}\bar{\mathcal{A}})^{\otimes n}$ carries the deconcatenation coproduct, with differential assembled from $\eta_{ij} = d \log(z_i - z_j)$ and the chiral product $\mu: j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \rightarrow \Delta_* \mathcal{A}$. The *modular convolution dg Lie algebra* is

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} := \mathrm{Hom}^{\mathrm{cyc}}(\bar{B}(\mathcal{A}), \mathcal{A}) \otimes \mathbb{G},$$

where $\mathbb{G} = \bigoplus_{g \geq 0} \mathbb{G}^{(g)}$ is the coefficient algebra generated by stable graphs on $\overline{\mathcal{M}}_{g,n}$. The genus filtration gives $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} = \bigoplus_{g \geq 0} (\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})_g$; the genus-zero part $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, (\circ)} = \mathrm{Hom}^{\mathrm{cyc}}(\bar{B}(\mathcal{A}), \mathcal{A}) \otimes \mathbb{G}^{(\circ)}$ is a strict dg Lie algebra with bracket $[-, -]^{(\circ)}$ from tree composition, since genus-zero stable graphs are trees.

2.2. The Maurer–Cartan element and degree truncations. The differential on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ splits as $D = d_o + D_{\mathrm{int}}$, with d_o linear and D_{int} the interacting piece encoding the chiral product and the Arnold form. The difference $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o \in \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is a Maurer–Cartan element [6, Thm. on MC2 bar-intrinsic]:

$$D\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

The element $\Theta_{\mathcal{A}}$ is bar-intrinsic. The bar complex is filtered by tensor degree, $F^r \bar{B}(\mathcal{A}) = \bigoplus_{n \leq r} (s^{-1} \tilde{\mathcal{A}})^{\otimes n}$, and the truncations $\Theta_{\mathcal{A}}^{\leq r} := \Theta_{\mathcal{A}} \bmod F^{r+1}$ form a tower in which each term is determined up to a coboundary by the previous one, via the Maurer–Cartan recursion.

Definition 2.1 (Shadow obstruction tower). The *degree- r shadow* of $\mathcal{A}$ is

$$\text{Sh}_r(\mathcal{A}) := -\nabla_H^{-1}([\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(o)}) \in \mathcal{A}_{r,o}^{\text{sh}}, \quad (2.1)$$

where the bracket is the genus-zero graph composition, the subscript r extracts the degree- r component, and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\text{sh}} = H^*(\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)})$ descended from the convolution Lie bracket. The sequence $(\text{Sh}_r(\mathcal{A}))_{r \geq 2}$ is the *shadow obstruction tower* of $\mathcal{A}$.

The definition is bar-intrinsic: if $\Theta_{\mathcal{A}} \equiv \Theta'_{\mathcal{A}} \bmod \text{gauge}$, then $\text{Sh}_r(\mathcal{A})$ and $\text{Sh}'_r(\mathcal{A})$ coincide as cohomology classes in $\mathcal{A}_{r,o}^{\text{sh}}$.

Remark 2.2 (Closed formulas). On the scalar lane, writing $\Theta_{\mathcal{A}} = \kappa \cdot \eta \otimes \Lambda$, one has $\text{Sh}_2 = \kappa$, $\text{Sh}_3 = \mathfrak{C}$, $\text{Sh}_4 = \mathfrak{Q}$, and the shadow metric identity $H(t)^2 = 4\kappa^2 + 12\kappa\mathfrak{C}t + (9\mathfrak{C}^2 + 16\kappa\mathfrak{Q})t^2 + \dots$ determines Sh_r for $r \geq 5$ from $(\kappa, \mathfrak{C}, \mathfrak{Q})$.

3. THE L_{∞} -FORMALITY OBSTRUCTION TOWER

3.1. Homotopy transfer on a strict model. Let $\mathfrak{g}$ be a strict dg Lie algebra with differential d and bracket $[-, -]$. A *deformation retract* onto $H^*(\mathfrak{g})$ is a triple (ι, π, h) with $\iota: H^*(\mathfrak{g}) \rightarrow \mathfrak{g}$, $\pi: \mathfrak{g} \rightarrow H^*(\mathfrak{g})$, and $h: \mathfrak{g} \rightarrow \mathfrak{g}$ of degree -1 , satisfying $\pi\iota = \text{id}$ and $\text{id} - \iota\pi = [d, h]$. The homotopy transfer theorem of Kadeishvili [2] and Kontsevich–Soibelman [4], in the form of Loday–Vallette [5, §10.3], produces a transferred L_{∞} -algebra structure $(H^*(\mathfrak{g}), \circ, \{\ell_n^{\text{tr}}\}_{n \geq 2})$ quasi-isomorphic to $\mathfrak{g}$, with brackets computed by the Kadeishvili–Merkulov tree sum

$$\ell_r^{\text{tr}}(a_1, \dots, a_r) = \sum_{T \in \text{Trees}_r} \frac{\pm 1}{|\text{Aut}(T)|} \pi \circ \Phi_T(\iota(a_1), \dots, \iota(a_r)), \quad (3.1)$$

where Trees_r is the set of planar rooted trees with r leaves and Φ_T decorates internal vertices with $[-, -]$ and internal edges with h .

3.2. The formality obstruction tower.

Definition 3.1 (L_{∞} -formality and obstructions). The dg Lie algebra $\mathfrak{g}$ is *L_{∞} -formal* if there is an L_{∞} quasi-isomorphism $\mathfrak{g} \simeq (H^*(\mathfrak{g}), \circ, [-, -])$; equivalently, $\ell_n^{\text{tr}} = 0$ in L_{∞} cohomology for all $n \geq 3$. The *L_{∞} -formality obstruction tower* is the sequence

$$(o_r(\mathfrak{g}))_{r \geq 3}, \quad o_r(\mathfrak{g}) := [\ell_r^{\text{tr}}] \in H^*(\text{Harr}^{\bullet}(\mathfrak{g})).$$

If $o_r(\mathfrak{g}) = 0$, an L_{∞} homotopy kills ℓ_r^{tr} and the next obstruction is defined.

For a pronilpotent dg Lie algebra with Maurer–Cartan element Θ , evaluating the transferred brackets on Θ produces a refined tower $(\ell_r^{\text{tr}}(\Theta, \dots, \Theta))_{r \geq 2}$, which takes values in the convolution target rather than in Harrison cohomology; this refined tower is the one compared with the shadow tower.

Remark 3.2 (Harrison equals cyclic at genus zero). For $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)}$, the Harrison complex controlling L_{∞} -formality coincides with the genus-zero cyclic deformation complex $\text{Def}_{\text{cyc}}^{(o)}(\mathcal{A})$, since genus-zero stable graphs are trees and the Harrison differential on trees is precisely operadic composition.

4. BASE CASES: DEGREES 2, 3, 4

Proposition 4.1 (Shadow–formality identification at low degree). *For every chiral algebra $\mathcal{A}$ the following identities hold:*

- (i) $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A}) = \ell_2^{(o)}(\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}).$
- (ii) $\text{Sh}_3(\mathcal{A}) = \mathfrak{C}(\mathcal{A}) = -h(\ell_3^{(o)}(\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 2})).$
- (iii) $\text{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A}) = [\ell_4^{(o), \text{tr}}(\Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 3})].$

Proof. *Part (i).* The binary bracket on $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}$ is the genus-zero convolution Lie bracket; evaluating on $\Theta_{\mathcal{A}}$ produces the scalar curvature $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ by definition [6, Thm. D]. The identity $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ is the degree-2 shadow on the scalar lane, so the three quantities agree.

Part (ii). The ternary bracket $\ell_3^{(o)}$, computed by (3.1), is a sum over $\text{Trees}_3 = \{T_s, T_t, T_u\}$, the three binary trees with three leaves. On the primary line $[\ell_3^{(o)}] = 0$ since $H^1(\overline{\mathcal{M}}_{0,4}) = H^1(\mathbb{P}^1) = 0$; at the chain level $\ell_3^{(o)}$ is exact and nonzero, and homotopy transfer gives $\mathfrak{C}(\mathcal{A}) = -h(\ell_3^{(o)}(\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 2})).$ The cubic shadow $\mathfrak{C}(\mathcal{A})$, computed as the three-channel graph sum at degree three, matches this three-tree sum because each binary tree with three leaves corresponds to one of the s -, t -, u -channel genus-zero stable graphs with four external legs, with identical weights and propagators.

Part (iii). The quaternary bracket $\ell_4^{(o)}$ is a sum over the 15 trivalent trees with five leaves, the faces of the Stasheff associahedron K_4 . Its transferred class $\ell_4^{(o), \text{tr}}$ lies in $H^*(\overline{\mathcal{M}}_{0,5})$. The weight-four graph sum defining $\mathfrak{Q}(\mathcal{A})$ enumerates precisely the same 15 trees with matching weights, signs, and propagators [6, App. on nonlinear modular shadows], so $\text{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A}) = [\ell_4^{(o), \text{tr}}(\Theta_{\mathcal{A}}^{\leq 3}, \dots, \Theta_{\mathcal{A}}^{\leq 3})].$ $\square$

Remark 4.2 (Degree-2: κ is family-dependent). The identity $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ holds at degree two on the scalar lane for every chiral algebra in the standard landscape; on the primary line this is the definition $S_2 := \kappa$. What is family-dependent is the value of κ itself: $\kappa^{\text{Vir}} = c/2$, $\kappa^{\text{Heis}} = k$, $\kappa^{\text{KM}} = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$, $\kappa^{\mathcal{W}_N} = c(H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$. The coincidence $\kappa = c/2$ holds only for Virasoro, not universally; writing $\kappa = c/2$ for a non-Virasoro family is a standard error.

5. THE MAIN THEOREM

Theorem 5.1 (Shadow obstruction tower equals formality obstruction tower). *For every chiral algebra $\mathcal{A}$ and every degree $r \geq 2$,*

$$\text{Sh}_r(\mathcal{A}) = \ell_r^{(o), \text{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \dots, \Theta_{\mathcal{A}}^{\leq r-1}), \quad (5.1)$$

where $\ell_r^{(o), \text{tr}}$ is the transferred genus-zero L_∞ bracket on the minimal model of $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}$. Consequently the shadow obstruction tower of $\mathcal{A}$ coincides with the L_∞ -formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}$ evaluated on the truncated Maurer–Cartan element.

Proof. Induction on the degree r . The base cases $r = 2, 3, 4$ are Proposition 4.1. The inductive step has four parts.

Step 1: Restriction to genus zero. Since genus-zero stable graphs are trees, $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}$ is a strict dg Lie algebra and the Harrison complex controlling L_∞ -formality coincides with the genus-zero cyclic deformation complex $\text{Def}_{\text{cyc}}^{(o)}(\mathcal{A})$. The Harrison differential on trees is precisely operadic composition on $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}$, so $\text{Harr}^\bullet(\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}) \xrightarrow{\sim} \mathcal{A}_{*, o}^{\text{sh}}$ is an isomorphism, not merely an injection.

Step 2: Homotopy transfer. Choose a deformation retract $(\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (o)}, h, \mathcal{A}_{*, o}^{\text{sh}})$ with $\pi \iota = \text{id}$, $\text{id} - \iota \pi = [D, h]$. The homotopy transfer theorem [2, 4, 5] produces a transferred L_∞ structure $(\mathcal{A}_{*, o}^{\text{sh}}, \{\ell_r^{(o), \text{tr}}\})$, with brackets computed by (3.1).

Step 3: Tree formula equals shadow graph sum. The degree- r shadow $\text{Sh}_r(\mathcal{A})$, by (2.1), expands $[\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(\circ)}$ as a sum over genus-zero stable graphs with r external legs, with internal vertices decorated by lower-degree shadows $\text{Sh}_j(\mathcal{A})$ ($j < r$) and internal edges by the propagator $P_{\mathcal{A}} = \nabla_H^{-1}$. These are the same trees and the same propagators as in (3.1): (a) a genus-zero stable graph with r external legs is a planar rooted tree $T \in \text{Trees}_r$ with valences determined by the graph; (b) the shadow propagator $P_{\mathcal{A}} = h \circ \pi$ corresponds to the homotopy h ; (c) the vertex evaluation $\text{Sh}_j = \pi \circ \ell_j^{(\circ)} \circ \iota^{\otimes j}$ corresponds to the vertex decoration $[-, -]^{(\circ)}$.

Step 4: Induction. At $r = 2$, both sides equal $\kappa(\mathcal{A})$ by Proposition 4.1(i); the cases $r = 3, 4$ are parts (ii), (iii) of the same proposition. Assuming (5.1) through degree $r - 1$, the graph sum for $\text{Sh}_r(\mathcal{A})$ and the tree formula for $\ell_r^{(\circ), \text{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \dots, \Theta_{\mathcal{A}}^{\leq r-1})$ range over identical planar rooted trees with identical internal-vertex decorations (via the inductive hypothesis), propagators, and signs. The two sums therefore agree as chains in $\mathcal{A}_{*,0}^{\text{sh}}$, completing the induction. $\square$

Remark 5.2 (Genus corrections decouple; uniform-weight scope). Theorem 5.1 is a genus-zero statement. On the uniform-weight scalar lane, the shadow obstruction tower $\text{Sh}_r(\mathcal{A})$ decomposes, at each genus $g \geq 0$ separately, as

$$\text{Sh}_r(\mathcal{A}) = \underbrace{\ell_r^{(\circ), \text{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \dots)}_{\text{genus-zero formality}} + \underbrace{\sum_{g \geq 1} \ell_r^{(g), \text{tr}}(\dots) + \Lambda_P(\dots)}_{\text{genus } \geq 1 \text{ corrections}},$$

where Λ_P is the BV loop operator. The classical and quantum contributions decouple at the level of the shadow decomposition: the quantum corrections form a parallel tower controlled by the quantum L_∞ structure, not by classical formality theory. At genus $g \geq 2$ with multi-weight generators, the scalar formula for the genus corrections receives a cross-channel term $\partial F_g^{\text{cross}}$; the genus-zero identification (5.1) nevertheless holds verbatim.

6. OPERADIC COMPLEXITY, PROVED

Definition 6.1 (Three complexity invariants). For a chiral algebra $\mathcal{A}$, set $r_{\max}(\mathcal{A}) := \sup\{r \geq 2 : \mathcal{A}_{r,0}^{\text{sh}} \neq 0\}$ (shadow termination degree), $d_\infty(\mathcal{A}) := \sup\{n \geq 2 : m_n^{\text{tr}} \neq 0\}$ (transferred A_∞ -depth on the minimal model of $\text{Def}_{\text{cyc}}(\mathcal{A})$), and $f_\infty(\mathcal{A}) := \sup\{n \geq 2 : \ell_n^{(\circ), \text{tr}} \neq 0\}$ (L_∞ -formality level of $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (\circ)}$).

Corollary 6.2 (Operadic complexity). $r_{\max}(\mathcal{A}) = d_\infty(\mathcal{A}) = f_\infty(\mathcal{A})$ for every chiral algebra $\mathcal{A}$.

Proof. The equality $r_{\max} = f_\infty$ is immediate from Theorem 5.1: $\text{Sh}_r(\mathcal{A}) = 0$ iff $\ell_r^{(\circ), \text{tr}}$ vanishes on the truncated Maurer–Cartan element. For $f_\infty = d_\infty$, the tree formula gives $\ell_r^{(\circ), \text{tr}} = \text{Alt}(m_r^{\text{tr}})$; since antisymmetrization is injective on cyclic cochains, $m_r^{\text{tr}} = 0$ iff $\ell_r^{(\circ), \text{tr}} = 0$. $\square$

Corollary 6.3 (Depth classification). *The standard landscape of chiral algebras is partitioned by the shadow depth into four classes, each realized by a standard family:*

$$\mathbf{G} (r_{\max} = 2), \quad \mathbf{L} (r_{\max} = 3), \quad \mathbf{C} (r_{\max} = 4), \quad \mathbf{M} (r_{\max} = \infty).$$

The class $\mathbf{G}$ consists of L_∞ -formal algebras; the class $\mathbf{M}$ consists of intrinsically non-formal algebras, in the sense that no finite truncation of their minimal L_∞ model is quasi-isomorphic to a dg Lie algebra.

Proof. Theorem 5.1 identifies shadow depth with L_∞ -formality level, and Corollary 6.2 identifies both with the transferred A_∞ -depth. The depth is therefore an invariant of the L_∞ quasi-isomorphism class of $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, (\circ)}$. Examples in Section 7 realize each class. $\square$

7. EXAMPLES

The four standard families realize the four shadow-depth classes.

Example 7.1 (Heisenberg; class **G**, formal). The Heisenberg vertex algebra Heis_k at level k , with $J(z)J(w) \sim k/(z-w)^2$, has $\kappa(\text{Heis}_k) = k$. The bar differential is strictly commutative, so $m_n = 0$ and $\ell_n^{(o),\text{tr}} = 0$ for $n \geq 3$. Hence $r_{\max}(\text{Heis}_k) = 2$ and Heis_k is L_∞ -formal. Class **G**.

Example 7.2 (Affine Kac–Moody; class **L**). The affine Kac–Moody vertex algebra $\widehat{\mathfrak{g}}_k$ has $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$. The binary bracket is the Lie bracket on $\mathfrak{g}$; Jacobi kills all degree-four graph contributions, so $\ell_4^{(o),\text{tr}} = 0$, while $\ell_3^{(o),\text{tr}} \neq 0$ in general. Hence $r_{\max}(\widehat{\mathfrak{g}}_k) = 3$. Class **L**.

Example 7.3 ($\beta\gamma$; class **C**, shadow depth 4). The rank-one bosonic ghost system $\beta\gamma$ has $r_{\max}(\beta\gamma) = 4$: the brackets $\ell_3^{(o),\text{tr}}$ and $\ell_4^{(o),\text{tr}}$ are nonzero, but rank-one rigidity kills degree-five graphs, giving $\ell_n^{(o),\text{tr}} = 0$ for $n \geq 5$. The primary invariants $(p_{\max}, k_{\max}, r_{\max}) = (1, 0, 4)$ are pairwise distinct: simple OPE pole, zero collision depth, shadow depth four. Class **C**.

Example 7.4 (Virasoro; class **M**, obstructed at every degree). The Virasoro vertex algebra Vir_c has $\kappa(\text{Vir}_c) = c/2$. The transferred brackets on the primary line of $H^*(\bar{B}(\text{Vir}_c))$ satisfy $m_k^{\text{tr}}(s^{-1}T, \dots, s^{-1}T) = S_k \cdot e_{2k}$ for $k \geq 2$, with $s^{-1}T$ the desuspension of the conformal vector to bar degree one and e_{2k} the weight- $2k$ basis vector of Ext^1 . The first shadow coefficients are

$$\begin{aligned} S_2 &= \frac{c}{2}, & S_3 &= 2 \text{ (} c\text{-independent)}, & S_4 &= \frac{10}{c(5c+22)}, \\ S_5 &= -\frac{48}{c^2(5c+22)}, & S_6 &= \frac{80(45c+193)}{3c^3(5c+22)^2}, & S_7 &= -\frac{2880(15c+61)}{7c^4(5c+22)^2}. \end{aligned}$$

All S_k are nonzero for generic c , so $r_{\max}(\text{Vir}_c) = \infty$: Virasoro is intrinsically non-formal. The Maurer–Cartan recursion forces every coefficient to be nonzero, since the normal-ordered product $:T^n:$ generates operations at every degree. Class **M**.

Remark 7.5 (The four classes exhaust the standard landscape). Classes **G**, **L**, **C**, **M** are the only possibilities consistent with the shadow metric identity $H(t)^2 = t^4 Q(t)$ with $\deg Q \leq 2$. A finite depth in $\{5, 6, \dots\}$ would force a higher-degree Q , excluded by the algebraicity theorem for the shadow generating function [6, Thm. on Riccati algebraicity]. Depth is therefore 2, 3, 4, or ∞ , and each is realized.

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